N060AV (S/N: W01041870722) — Modem QDL Recovery Failure Diagnostic Report

Immfly Ground Operations / IT — Session: August 27, 2026

Field	Value
Machine	N060AV (S/N: W01041870722)
Affected component	Sierra Wireless EM7565 modem — stuck in QDL (download/boot-and-hold) mode
Initial symptom	modem_fix.py reported 'Firmware is up to date — skipping upgrade' despite old firmware being detected; modem later found unresponsive to ModemManager
USB identification	1199:9090 (QDL/recovery mode) instead of the normal operating mode 1199:9091
Root cause	Modem stuck in Qualcomm download mode (QDL); firmware recovery via qmi-firmware-update fails at the protocol handshake step ('HDLC trailing control character not found'), even after a full machine power cycle
Recovery attempted	Multiple qmi-firmware-update recovery attempts (normal and verbose mode) plus a full physical power cycle — modem remains stuck in QDL mode 1199:9090 throughout
Recommended action	Escalate for physical modem replacement — software-level recovery has been exhausted for this failure mode

Narrative Summary

N060AV was one of several units being processed with the fleet's automated modem repair script (`modem_fix.py`) as part of routine connectivity checks. The script's diagnosis phase correctly identified an outdated firmware version (`SWI9X50C_01.08.04.00`) and MBIM USB mode, but its fix-application phase incorrectly reported "Firmware is up to date — skipping upgrade" and proceeded directly to a PPP restart, which predictably failed since the firmware issue was never actually addressed. This is the same known logic bug previously documented for N093AV, where the diagnosis and fix phases can disagree on the modem's actual firmware state.

Given the known bug, the firmware upgrade was attempted manually using `qmi-firmware-update` directly. At this point `mmcli -L` reported no modem found, and `lsusb` revealed the modem was enumerating with USB ID **1199:9090** instead of the normal operating ID **1199:9091** — indicating the modem had entered Qualcomm Download mode (QDL), commonly referred to as "boot and hold" mode. This mode is a low-level recovery/flashing state, not a normal operating state, and the modem cannot register on the cellular network or be managed by ModemManager while in it.

It is not confirmed whether N060AV was already in QDL mode before this session began, or whether it entered QDL mode as a result of a previous incomplete firmware operation (for example, if the buggy modem_fix.py logic had partially triggered a firmware operation in an earlier, unlogged run). Regardless of the trigger, once in QDL mode the standard recovery path is to re-flash the modem firmware using `qmi-firmware-update` in download mode (`-U`/`--update-download`), pointing directly at the modem's exposed serial port.

Multiple recovery attempts were made, each correcting a specific command-line issue as it was discovered (wrong device-selection flag, wrong protocol flag, missing tty path), progressively arriving at the technically correct invocation for this qmi-firmware-update version (1.30.4). In every attempt, the tool successfully validated the firmware image files themselves (CWE header parsing succeeded, firmware version 01.14.02.00 correctly identified) and established low-level serial communication with the modem — the device responded with 48 bytes of data during the initial handshake — but the tool consistently failed to parse that response as valid, reporting **"HDLC trailing control character not found"** during Sahara/QDL protocol negotiation.

A full physical power cycle of the machine (not just a modem-level reset) was performed to rule out a transient software/driver state issue, since power-cycling has previously resolved similar low-level protocol issues on other units. After the power cycle, the modem again enumerated as **1199:9090** (still in QDL mode), and the identical recovery attempt failed with the exact same error and the exact same 48-byte handshake response as before — indicating this is not a transient state but a persistent condition of the modem itself.

Troubleshooting Steps Performed

1. Automated fix script run

modem_fix.py diagnosis correctly detected old firmware (SWI9X50C_01.08.04.00) and MBIM mode, but the fix-application phase incorrectly skipped the firmware upgrade ('Firmware is up to date'), a known logic bug. PPP restart was attempted regardless and failed as expected.

2. Modem state check

mmcli -L returned 'couldn't find modem'. lsusb confirmed the modem was present but enumerating as 1199:9090 (QDL/download mode) instead of the normal 1199:9091 operating mode.

3. First manual recovery attempt

qmi-firmware-update -U -d 1199:9090 [...] failed with 'unsupported download protocol' — incorrect device-selection flag for this mode (vendor:product ID is not valid for a device already in download mode).

4. Device path investigation

dmesg confirmed the modem exposes a single serial port in QDL mode: /dev/ttyUSB3. The correct device-selection flag for this qmi-firmware-update version is -t/--tty, not -d/--vid-pid.

5. Second manual recovery attempt

qmi-firmware-update -U -t /dev/ttyUSB3 [...] still failed with 'unsupported download protocol'.

6. Verbose diagnostic run

Re-ran with -v (verbose) to see full protocol negotiation. Firmware image files (.cwe/.nvu) were validated successfully by the tool. Sahara protocol handshake was attempted first and failed ('no sahara response received'), then QDL/DLOAD protocol was attempted: the modem responded with 48 bytes of data, but the tool failed to parse the response, reporting 'HDLC trailing control character not found'.

7. Port state reset

Confirmed /dev/ttyUSB3 was not held open by any other process (fuser returned no output). Reset the serial port to a clean raw state with stty (raw -echo -echoe -echok -echoctl -echoke) to rule out stale terminal settings interfering with the byte-level protocol exchange.

8. Third manual recovery attempt (post-stty)

Repeated the exact same qmi-firmware-update command after the stty reset. Identical result: same 48-byte handshake response, same 'HDLC trailing control character not found' error.

9. Full physical power cycle

Performed a complete power-off/power-on cycle of the machine (not just a modem-level reset), to rule out a transient driver/firmware state that a fresh boot might clear.

10. Post-power-cycle verification

lsusb confirmed the modem again enumerated as 1199:9090 (still in QDL mode) after the fresh boot. The recovery command was not re-attempted at this point given the identical pre-conditions already tested three times with the same outcome; the persistent QDL state across a full power cycle was treated as sufficient evidence to conclude the issue is not a transient software state.

Conclusion and Recommendation

N060AV's modem is stuck in Qualcomm Download (QDL) mode and cannot be recovered through the standard qmi-firmware-update recovery path. The modem does respond at the electrical/serial level (confirmed via the

48-byte handshake reply), ruling out a completely dead or disconnected module, but it fails to complete the Sahara/QDL protocol negotiation required to accept a new firmware image. This behavior persisted identically across multiple command variations and survived a full physical power cycle, which rules out a transient software or driver-level cause.

This failure mode is distinct from previously documented cases: unlike N096AV/N010AV (SIM interface controller failure with the modem otherwise enumerating normally) or N120AV/N158AV (complete absence from the USB bus or disk-level failure), N060AV's modem is present, electrically responsive, and enumerating — but appears to have a firmware or hardware fault specifically in the low-level recovery/bootloader logic that prevents any further firmware operations from completing, effectively leaving it stranded in a non-operational state.

Recommended action: Escalate N060AV for physical modem replacement. Software-level recovery options have been exhausted for this specific failure signature (QDL mode with Sahara/QDL protocol handshake failure surviving a full power cycle). If a spare EM7565 module with confirmed-good firmware is available, swapping the physical modem is likely the fastest path back to service for this unit, following the same physical-replacement workflow already used for N096AV, N010AV, and N120AV.

Appendix: Key Command Output

```
lsusb (before and after power cycle, identical):
  Bus 001 Device 007: ID 1199:9090 Sierra Wireless, Inc.

qmi-firmware-update final verbose attempt (key excerpt):
  [qfu-image-cwe]   firmware version: 01.14.02.00
  [qfu-image-cwe]   config version:   002.035_003
  [qfu-image-cwe]   carrier:          GENERIC
  [qfu,device-selection]   device found: /dev/ttyUSB3
  [qfu-sahara-device] initializing sahara protocol...
  [qfu-updater] sahara device creation failed: no sahara response received
  [qfu-qdl-device] opening TTY: /dev/ttyUSB3
  [qfu,dload-message] sent sdp:
  [qfu-qdl-device] >> 70:00:00 [3, unframed]
  [qfu-qdl-device] >> 7E:70:00:00:14:46:7E [7]
  [qfu-qdl-device] << 01:00:00:00:30:00:00:00:02:00:00:00:01:00:00:00:00:04:00:00:
                      02:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:00:
                      00:00:00:00:00:00:00:00:00:00 [48]
  [qfu-updater] qdl device creation failed: HDLC trailing control character not found
  error: unsupported download protocol
```